Fundamentals of Affective Computing

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A Definition

The field of affective computing encompasses both the creation of and interaction with machine systems that sense, recognize, respond to, and influence emotions (Picard, 1997; Picard and Klein, 2002).

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HAL: Do you read me?

2001: A Space Odyssey: A Conversation with HAL (1968)

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Course Topics

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An exciting vision of AI

- Speech,
- Facial recognition,
- Lip reading,
- Systems monitoring,
- Regulation and control,
- Game play, and
- Social skills

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Affect Sensing

- Affect sensing refers to a system that can recognize emotion by receiving data through signals and patterns (Picard, 1997).

- To accomplish this task, a computer would need to be equipped with hardware and software.

- Affect-sensing systems can be classified by modalities, each of which has a unique signature.

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Affect Sensing

- Affect-sensing systems can be classified by modalities:
  - Facial activity,
  - Posture, Gesture,
  - Hand tension,
  - Vocal,
  - Textual,
  - Physiological Signals
    - Electrodermal activity (EDA),
    - Electroencephalography (EEG) etc.

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Applications (Picard , 1997)

1. **Detection:** Systems that detect the emotions of the user

2. **Expression:** Systems that express what a human would perceive as an emotion (e.g., an avatar, robot, and animated conversational agent)

3. P**erception**: Systems that actually 'feel' an emotion."

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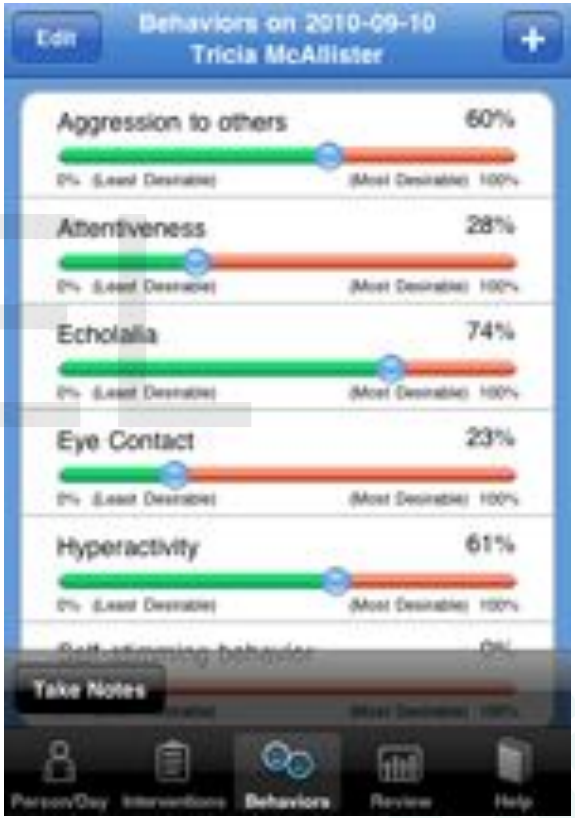
Health Care Applications

- Individuals with Asperger syndrome (AS) or high functioning autism (HFA)
  - Mobile applications like SymTrend and Autism Track (HandHold Adaptive)



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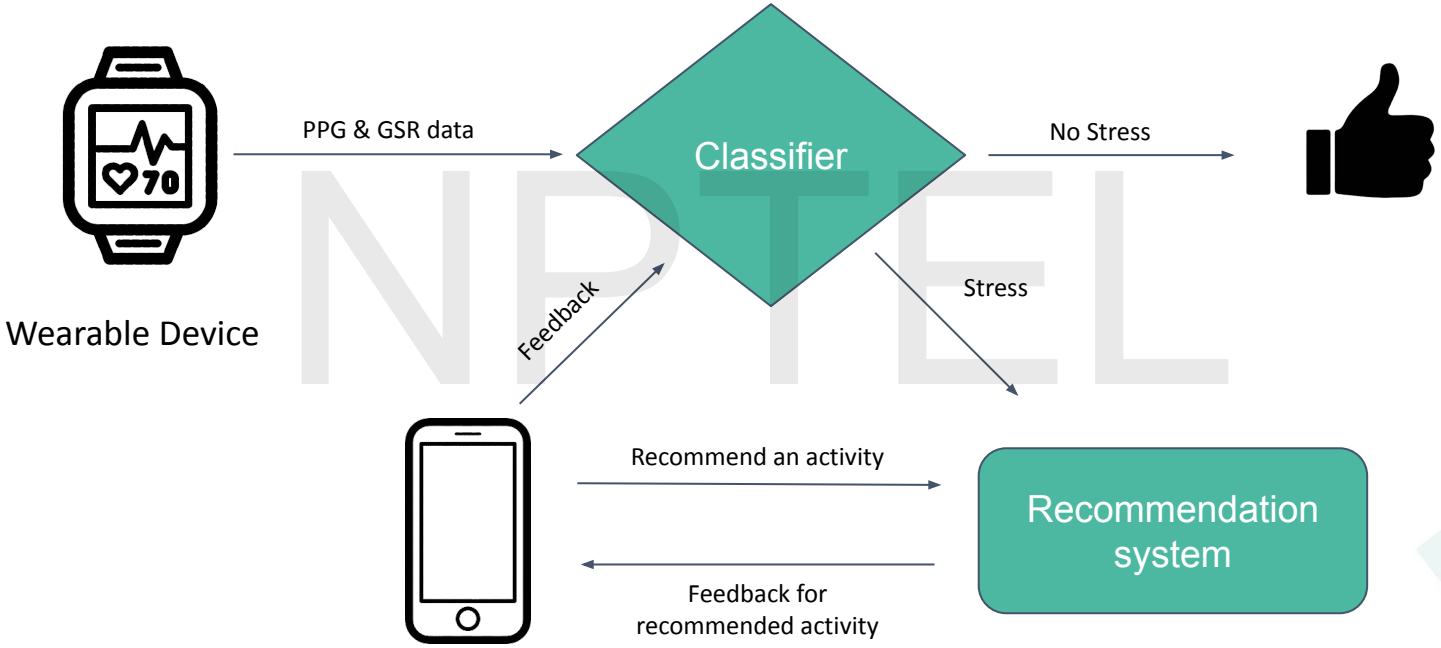
HealthCare Applications: Visharanti



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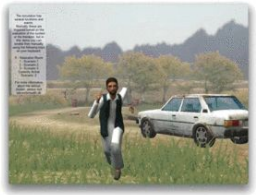
Health Care Applications

- Post-traumatic stress disorder (PTSD)
  - StartleMart,
  - Virtual Vietnam, Virtual Iraq, Virtual Afghanistan

(a) Sound of ventilator blowing overhead.

(b) Sound of wind blowing.

(c) Man walking angrily toward player.



(d) Afghan running toward player.

(e) Man staring at player.

(f) Wounded soldier staring at player.

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Education Applications

- EngageMe (Darnell, 2014).
  - This system uses skin conductance data collected from students in a classroom, along with video feeds, to help the teacher reflect on his or her classes.
- Subtle Stone (Balaam, 2009)
  - Wireless, hand-held, squeezable ball that allows students to communicate their affective and motivational experiences to their teachers in real time.
- China Wants to Raise Smarter Students

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Other: Preparing you for interview

- MACH (My Automated Conversation coacH)

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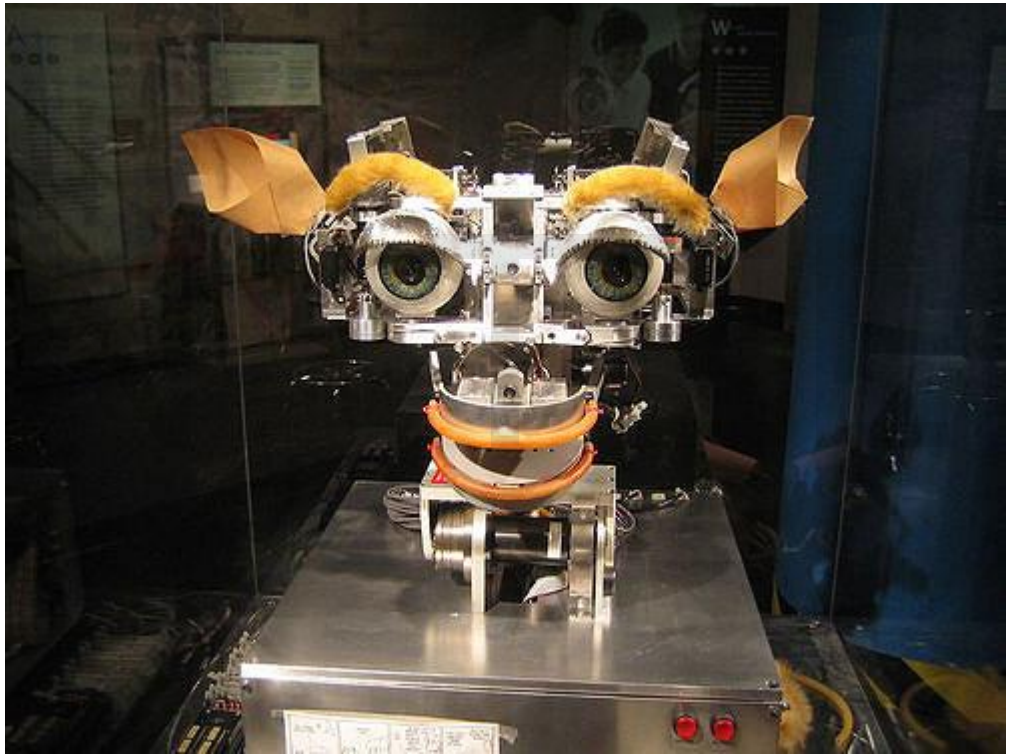

Case Study: It’s all about Experience

It's all about User Experience

- UX is a broad concept that encompasses many aspects of a user's interaction with your product and company; namely "how they feel about you."
  - Marketing Strategy by Steve Jobs
- Usability is more the practical side of things, the main question being answered is "ease of use."
- The combination of the two leads to a better understanding of emotional experience and interaction with your brand and hopefully adds to a memorable, positive experience.
  - *But, how do you test for both?*

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Participatory Design/Acceptance Test

- Participatory design is a process that includes the stakeholders in the early stages of design.
  - Participatory design lends itself to user-centered design innovation.
    - PICTIVE approach (Muller, 1991)
- Acceptance Testing
  - During Beta testing the end-user (intended real user) validates the product for functionality, usability, reliability, and compatibility.
    - Self-reporting in form of focus groups/ surveys

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Problems!

- Users have to articulate exactly how they feel about the products in the form of focus groups and surveys, which can be plagued by issues like participant bias, recall bias etc.
- Advanced UX/Neuromarketing
  - Application of Affective Computing/Neuroscience to marketing.
    - How customers use their mind while responding to products?
    - How this impacts their decision making?
- *To gain new perspectives and have a more accurate measure of customers' functional mind*

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Advanced UX/Neuromarketing

- For the purpose of this class, we will focus on web-based interactions
  - Some main considerations when embarking on advanced UX & usability testing with Affective Computing.

- The primary elements of our case study will focus on how to employ Affective Computing techniques for web-based testing.

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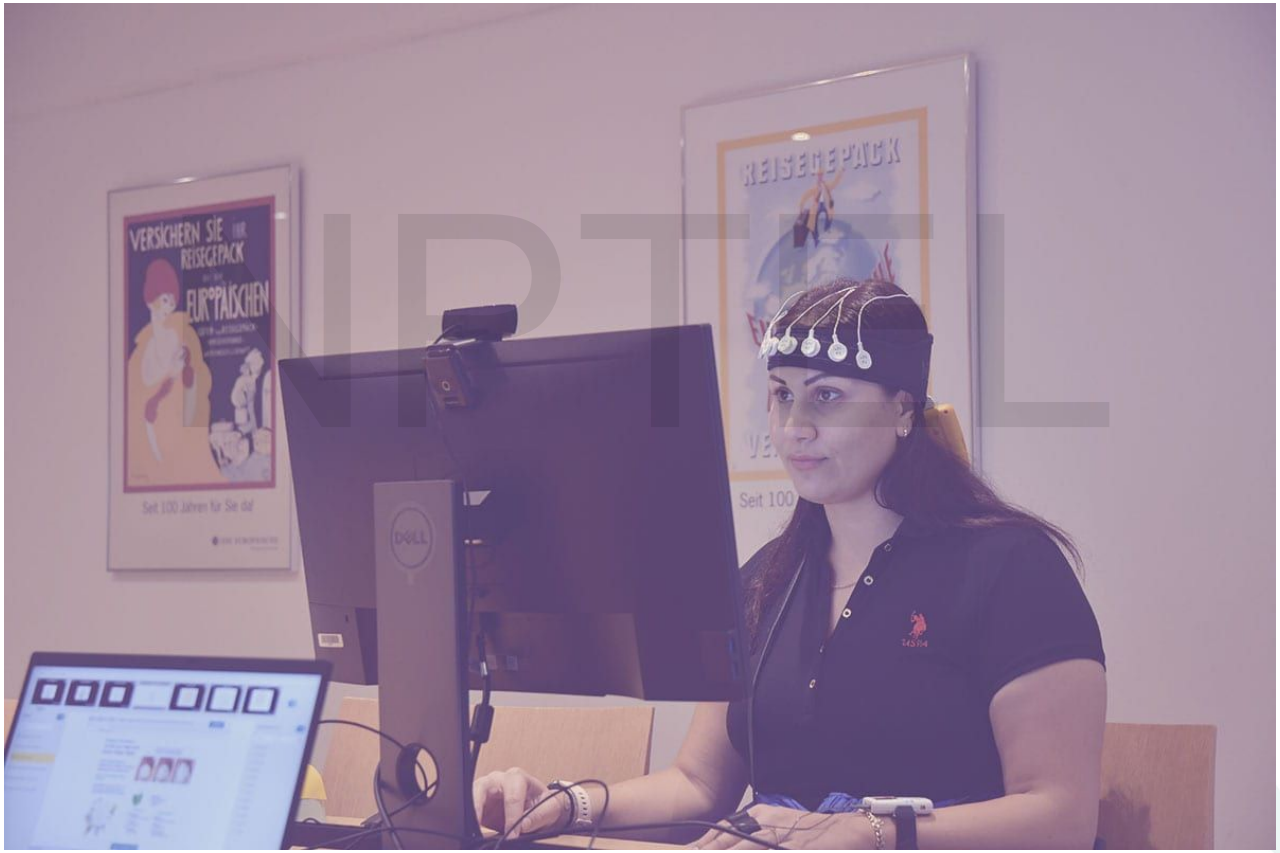
Advanced UX/Neuromarketing



Ethical Considerations

- **Emotional Manipulation:** Is it ethical for computers to detect, recognize, and then attempt to modify certain behaviors?

- **Privacy:** Emotions, perhaps more so than thoughts, are ultimately personal and private (Picard, 2003).

- **Emotional Dependency:** Could having users routinely use these *moral agents* create codependence?

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